

FINDING SIMILAR ITEMS

Applications of Set Similarity
and Shingling of Documents

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SUMMARY

- Introduction
- Applications of Set Similarity
- Shingling of Documents
- Conclusion



INTRODUCTION

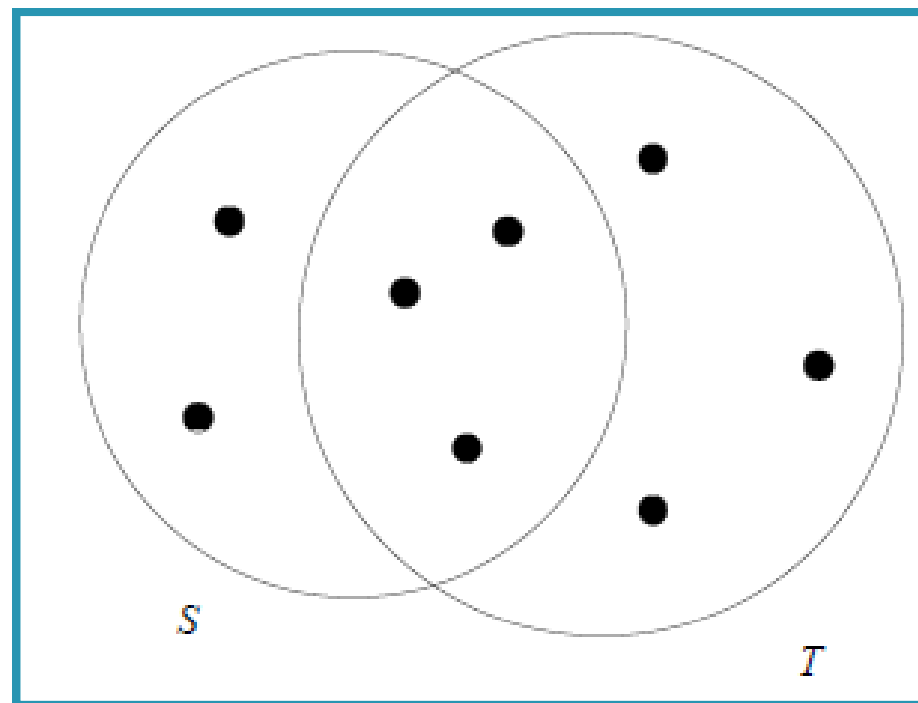
A fundamental data-mining problem is to examine data for “similar” items.

It is usefull to identify plagiarism cases or to recommend articles to customers.



APPLICATION OF SET SIMILARITY

Jaccard Similarity :
$$\text{SIM}(S,T) = \frac{|S \cap T|}{|S \cup T|}$$



APPLICATION OF SET SIMILARITY

Plagiarism :

A comparing process word by word will probably not detect a sophisticated plagiarism. A plagiarizer may extract only some parts of a document and alter few words and the sentences order.

Mirror Pages :

Web sites are commonly duplicating pages to share the load. These mirror sites will be quite similar but rarely identical. It is important to find these pages to avoid showing all of them within the first page of results.

Articles from a same source :

A same article can be distributed in few web sites and newspapers. The core of each page will be the same as the original article, but the content may differ.



APPLICATION OF SET SIMILARITY

This notion of similarity we saw is only looking to textually similar documents, **but not the “similar meaning”**.



APPLICATION OF SET SIMILARITY

Collaborative filtering :

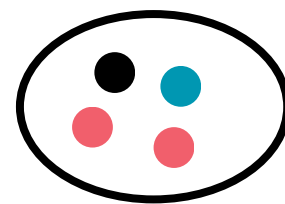
Recommendation to users items that were liked by other users with similar tastes

amazon

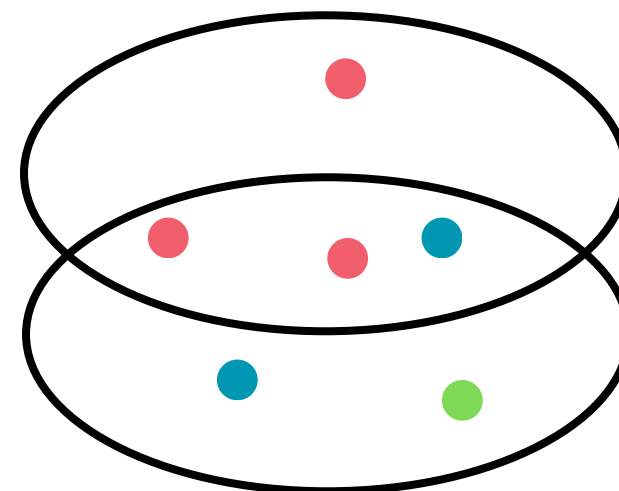
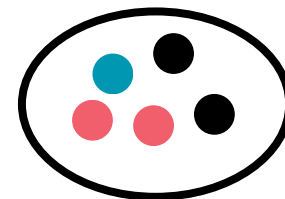
NETFLIX

Example :

Set A
 $\{\underline{a}, \underline{a}, \underline{a}, \underline{b}\}$



Set B
 $\{\underline{a}, \underline{a}, \underline{b}, \underline{b}, \underline{c}\}$



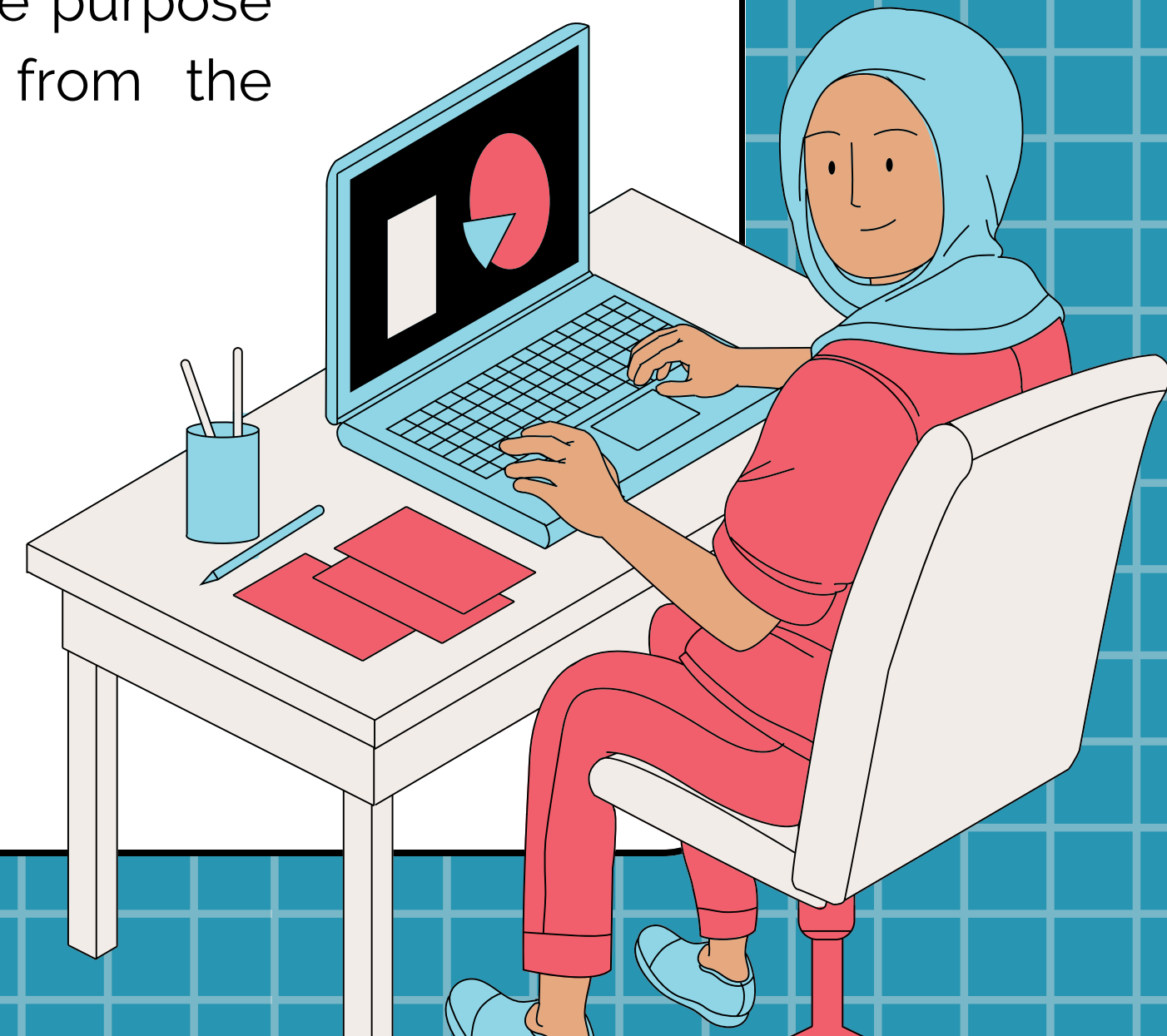
Jaccard Similarity

$$3/6 = 0.5$$

SHINGLING OF DOCUMENTS

The most effective way to represent documents as sets for the purpose of identifying lexically similar documents is to construct from the document the set of short strings that appear within it.

If we do so, then documents that share pieces as short as sentences or even phrases will have many common elements in their sets, even if those sentences appear in different orders in the two documents.



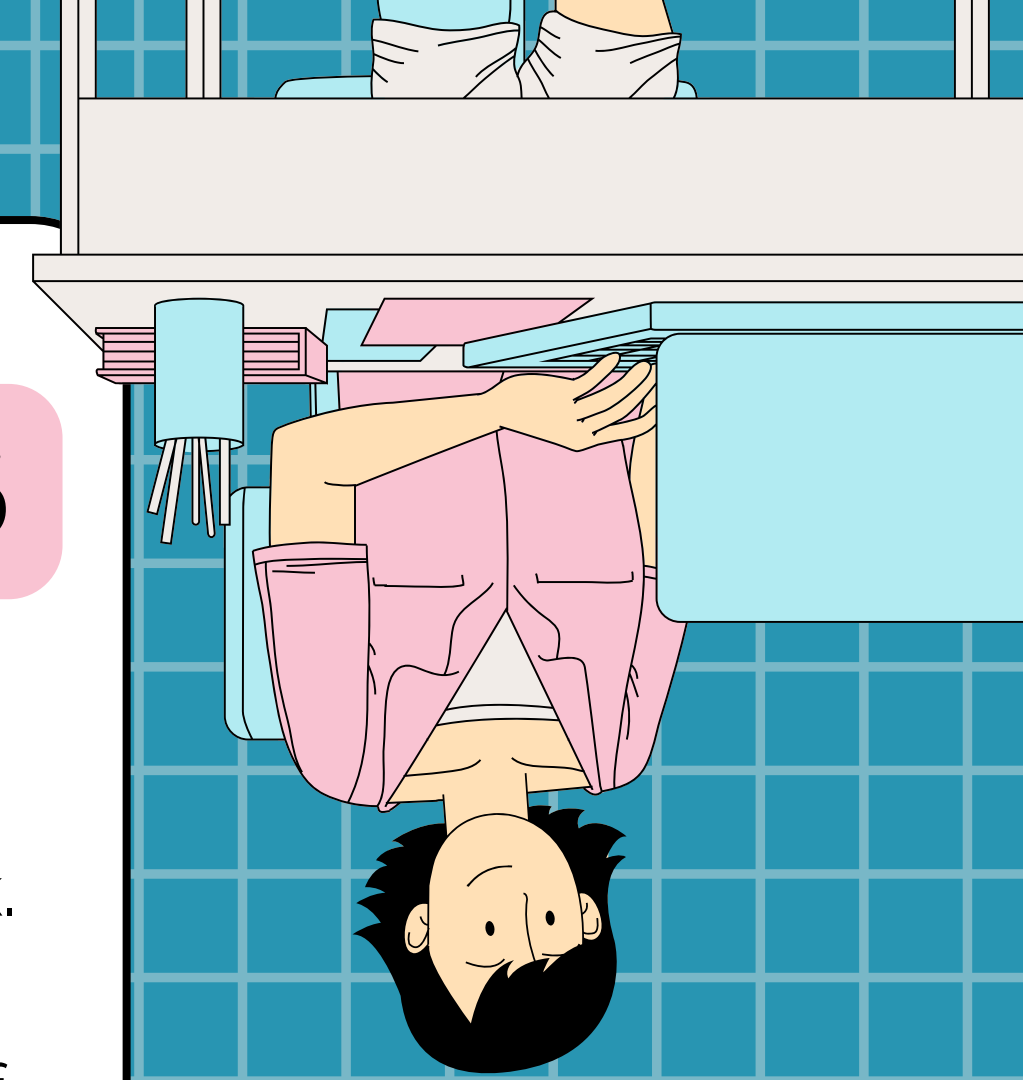
SHINGLING OF DOCUMENTS

A document is a string of characters, we define a **k-shingle** for a document to be any substring of length k .

Thus, we can associate with each document the set of k -shingles that appear one or more time within a document.

Example : Our document D is : "abcdef" and $k=2$

The set of 2-shingles will be : {ab, bc, cd, de, ef}

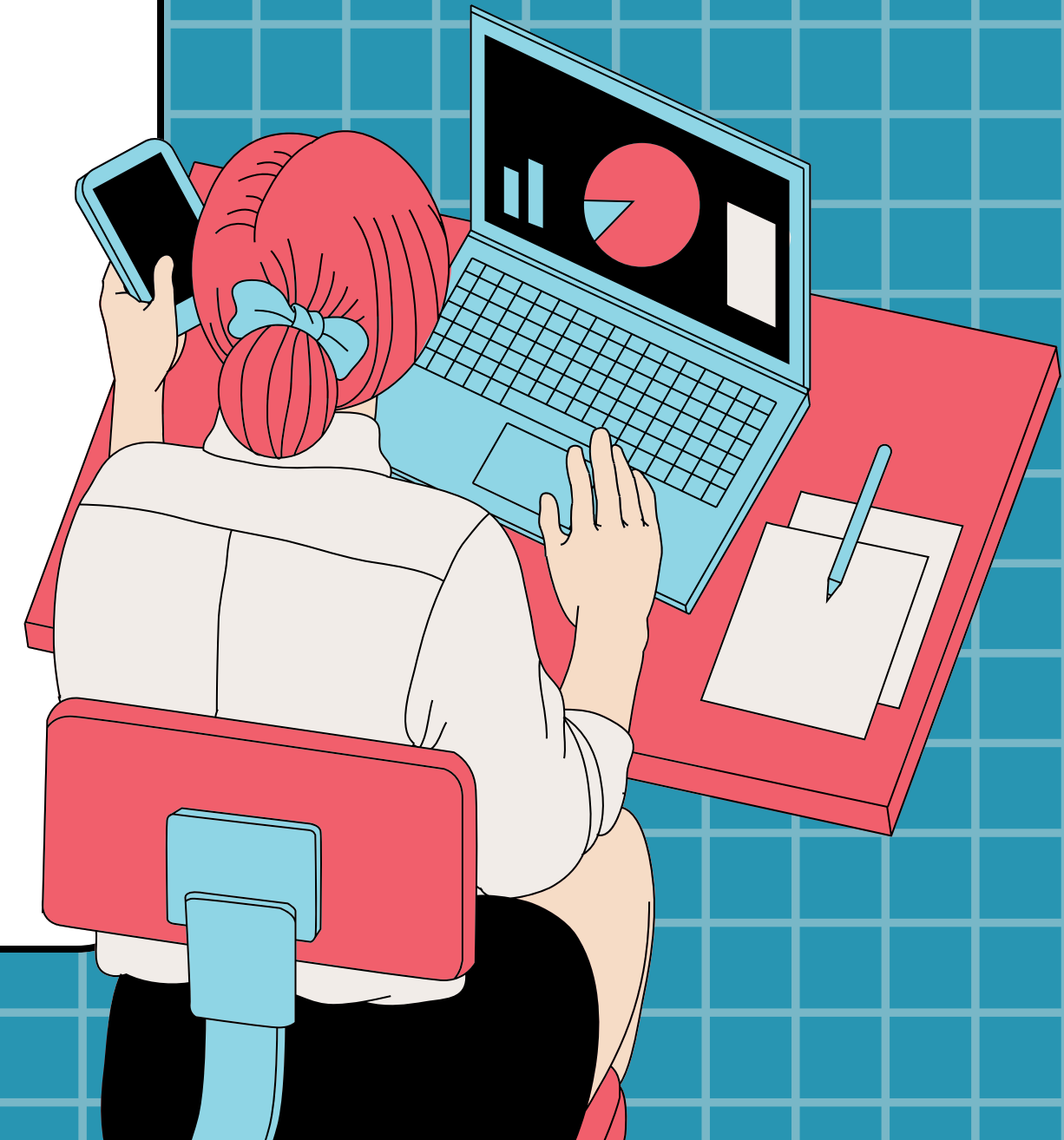


SHINGLING OF DOCUMENTS

Picking a **k too small is not a good idea** because all the documents will be similar according to the Jaccard Similarity.

k should be large enough that the probability of any given shingle appearing in any given document is low.

Considering that only 20 letters are commonly appearing in documents, we will be estimating the number of k-shingles as 20^k . **k=9 is considered a safe choice.**





SHINGLING OF DOCUMENTS

Instead of using substrings as shingles, we can pick a **hash functions** that maps strings of length k to some number of buckets and treat the resulting bucket number as the shingle.

The set representing a document is then the set of integers that are bucket numbers of one or more k -shingles that appear in the document.

We could construct the set of 9-shingles for a document and then map each of those 9-shingles to a bucket number in a range of 0 to $2^{32} - 1$. Which mean that each shingle can be represented by 4 bytes instead of 9.

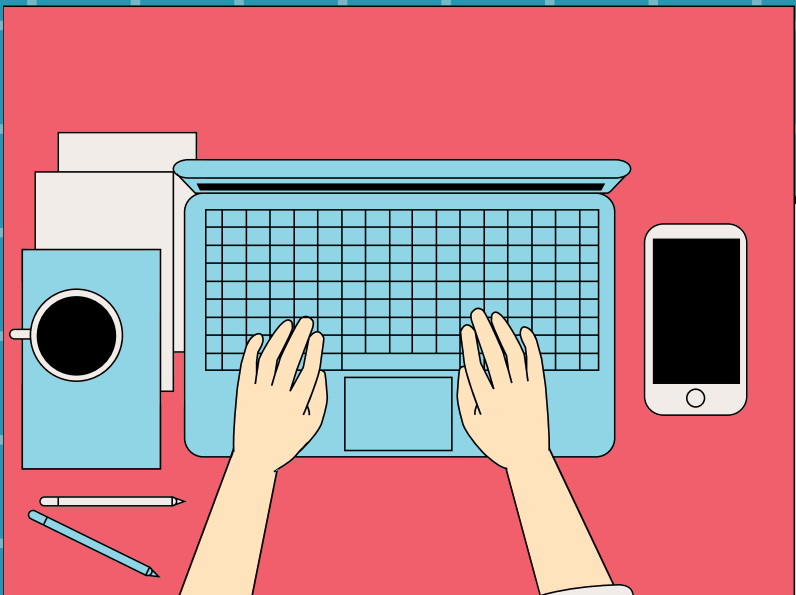
SHINGLING OF DOCUMENTS

The documents are commonly using a lot of “**stop words**”, such as “and”, “you”, “to”...

They are not really usefull to identify topics and similarities, thus we should ignore them.

However, it was found that defining a shingle to be a **stop word followed by the next two words**, can **form a usefull set of shingles**.

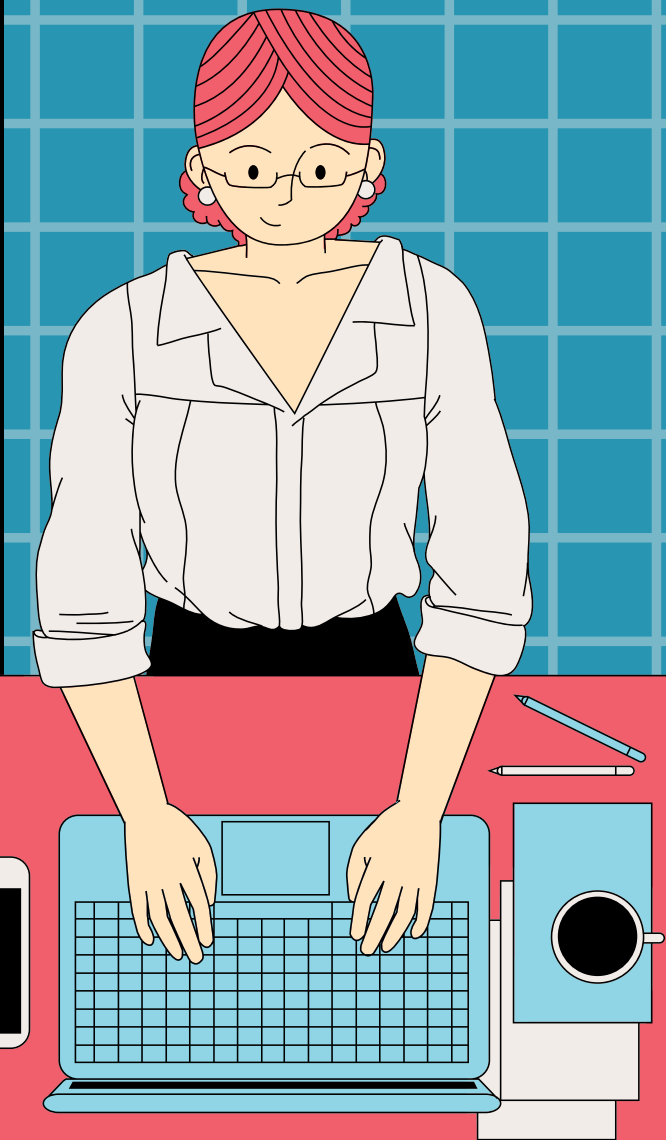




CONCLUSION

Jaccard Similarity : Ratio of the intersection of the sets to the size of the union. This measure of similarity is effective for many applications, such as textual similarity of documents and similarity of buying habits of customers.

Shingling : A k-shingle is any k characters that appear consecutively in a document. By representing a document by its k-shingles, we can use the Jaccard Similarity to observe textual similarities. We can hash shingles to bit strings of shorter length.



**ARE THERE ANY
QUESTIONS?**

Thank for your time

